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Paleomagnetism of the Upper Cretaceous Mount Somers Volcanics, Canterbury, New Zealand

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ABSTRACT

The Mount Somers Volcanics of mid Canterbury in eastern South Island are a high-K calc-alkaline andesite–dacite–rhyolite suite erupted subaerially approximately 95 Ma ago during the long Late Cretaceous episode of normal geomagnetic field. Thermoremanent magnetic directions at 46 sites, corrected for post-eruption tectonic tilt, are stable and group well ($\alpha_{95} = 3{\cdot}8°$, $K = 31{\cdot}7$) with a very steep mean inclination ($I = -85°$, $D = 354°$). The resulting paleomagnetic South Pole position is at 52° S 174° E, indicating that Canterbury was within 10° of the South Pole 95 Ma ago, and that little finite rotation of Canterbury relative to the South Pole has occurred since.

The Mount Somers Volcanics were erupted prior to the separation of eastern New Zealand from West Antarctica along the Pacific-Antarctic ridge. They provide a mid-Cretaceous pole position that can be compared with pole positions of similar age from Australia and with the polar-wander path previously reported for the Chatham Islands over the past 75 Ma. The Mount Somers pole fits neatly onto the New Zealand polar-wander path, which is consistent with the Australian polar-wander path when convergent displacements across the Pacific-Indian plate boundary, and spreading of the Tasman Sea, are allowed for.

The paleomagnetic data also support the proposition that regional folding and oroclinal bending of the pre-Cretaceous Torlesse basement rocks mainly predates the extrusion of the Mount Somers Volcanics and took place primarily during the Early Cretaceous Rangitata Orogeny and not during the Cenozoic as implied in recent plate-tectonic reconstructions.

INTRODUCTION

Paleomagnetic pole positions determined from some eastern South Island volcanic rocks, using natural remanent magnetisation (NRM) data, gave no indication of polar wandering, with respect to that part of New Zealand east of the Alpine Fault, since the Eocene period (Coombs & Hatherton 1959). Paleomagnetic pole positions determined from Chatham Island volcanics, using results from cleaned samples after comprehensive stability tests, indicated no significant differences from the present pole position at 5 Ma B.P. and 35–40 Ma B.P., but a position 19° of latitude further south at 75–80 Ma B.P. (Grindley *et al.* 1977). These results also indicated no independent rotation of the Chatham Rise relative to the mainland of New Zealand.

The Mount Somers Volcanics of early Late Cretaceous age (90–95 Ma B.P.), situated in mid Canterbury on the eastern margin of the Southern Alps (at latitude 43° 40′ S and longitude 171° 25′ E) and only 70–80 km from the Alpine Fault, provide a further means of testing the latitudinal displacement of eastern New Zealand during Late Cretaceous sea-floor spreading from the Pacific-Antarctic ridge, and a possible check on the timing of dextral shear on the Alpine Fault and oroclinal bending of the Mesozoic basement rocks of the New Zealand Geosyncline (Suggate 1963; Grindley 1963; Fleming 1970). The Mount Somers Volcanics, being 20 Ma older than the Chatham Islands Late Cretaceous volcanics, also provide additional information on the apparent polar-wander path of eastern New Zealand to compare with that of Australia.

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GEOLOGICAL OUTLINE

The Upper Cretaceous terrestrial calc-alkaline Mount Somers Volcanics of Canterbury Province, crop out discontinuously for 70 km along the foothills of the Southern Alps (Fig. 1). The volcanics unconformably overlie Jurassic and Triassic Torlesse Supergroup sedimentary rocks and, at places, are overlain by Lower Tertiary (Paleocene) coal measures and silica sands. They crop out in fault-bounded blocks upthrown to the north-west during the Late Tertiary–Recent Kaikoura Orogeny.

The volcanic rocks are high-alumina basalt, dolerite, andesite, dacite, and rhyolite, which have been dated at about 95 Ma B.P. by the K-Ar method on total rock, sanidine, and plagioclase from 24 samples representative of the rock-types. The age range determined for these volcanics was from 98 Ma B.P. to 92 Ma B.P. (Oliver 1977; Adams & Oliver in press). The rhyolitic rocks mainly form domes, flows, ignimbrites, and water-laid tuff beds; while the intermediate rocks are mainly flows and interlayered tuffs.

Petrographically, the rhyolitic rocks are composed of quartz, sanidine, and plagioclase phenocrysts, minor to rare almandine garnet, and biotite, and a finely grained felsitic groundmass. Secondary hematite is present in some of the rhyolites. The dacite, andesite, and high-alumina basalt consist of plagioclase and hypersthene phenocrysts, minor ilmenite, magnetite, and clinopyroxene, in a hyalopilitic to interstitial groundmass. Garnetiferous xenoliths of metasedimentary origin are

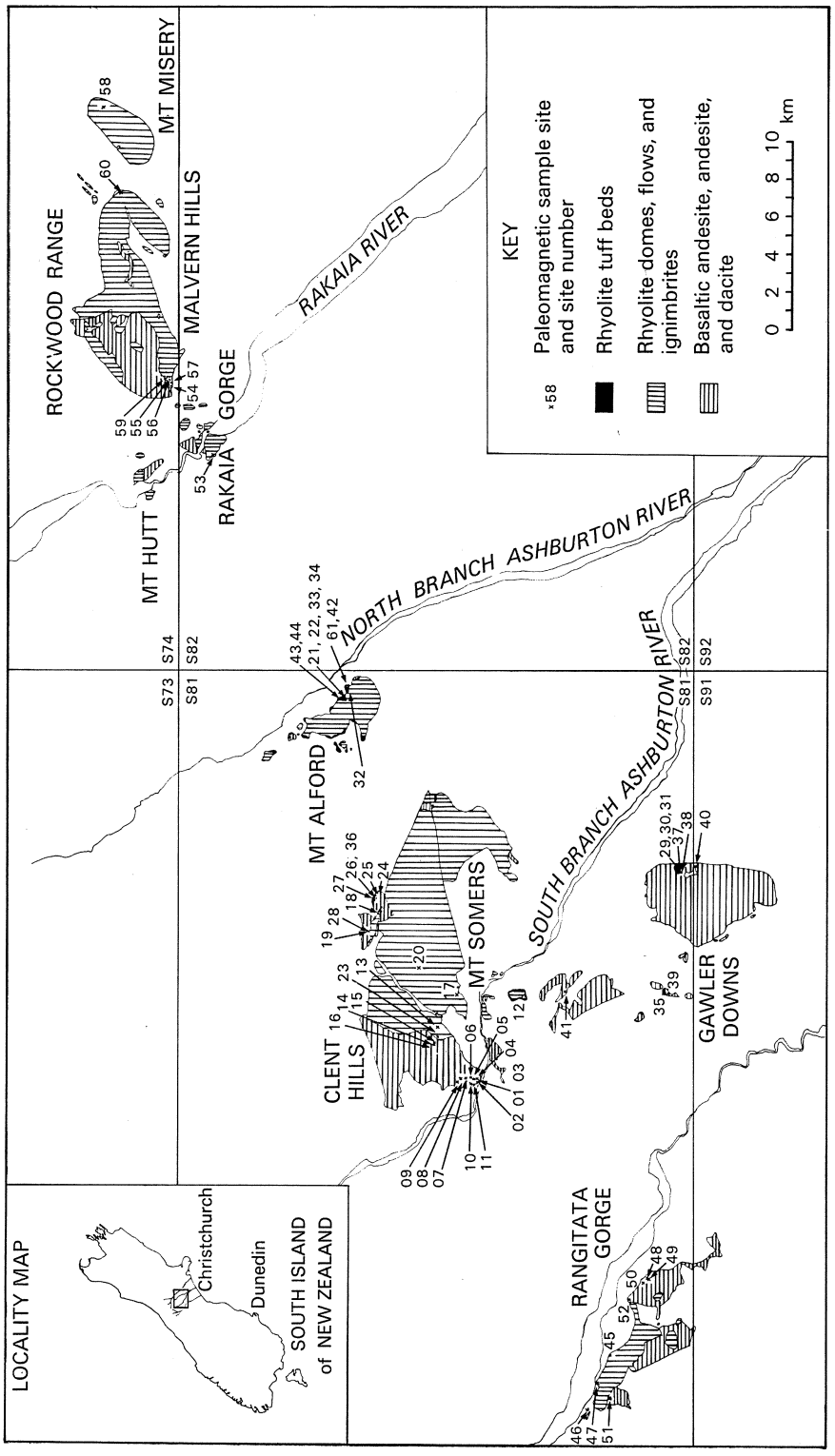


FIG. 1—Locality map of the Mt Somers Volcanics and paleomagnetic sampling sites.

ubiquitous in the dacite, but are absent from the andesite and basalt.

All the volcanic rocks are geochemically interrelated and apparently form a continuous differentiation sequence (Oliver 1977) that erupted over a short period of time (probably less than 5 Ma).

PALEOMAGNETIC MEASUREMENTS

Paleomagnetic samples of Mount Somers Volcanics were collected only from fresh rocks to avoid secondary remanent magnetisation caused by alteration and weathering. Exposures of fresh andesite and dacite are abundant, but the rhyolite is generally altered along well-developed closely spaced joints.

The samples were collected from sites previously sampled for radiometric dating, and also from sequences of flows, to average out secular variation.

Specimens measured were cylindrical, 25 mm in diameter, and 25 mm long, cut from cores of various lengths, some of which were taken in the field with a portable drill (Doell & Cox 1965; 1967), and others drilled in the laboratory from oriented blocks. Each core is referred to as a sample (N, see Table 2), most samples consisting of two to four specimens. Each site consisted of a single flow, dome, ignimbrite sheet, or bed, representing a single point in time. Details of the paleomagnetic collection sites are given in Table 1 and Fig. 1.

The core specimens were measured on one of three types of magnetometer. The weakest specimens (mainly rhyolitic tuffs and rhyolites) were measured on a Schonstedt-type spinner magnetometer connected to a P.A.R. Model 129 two phase/vector lock-in amplifier. Some samples with a higher intensity were measured on a 5 Hz Fluxgate spinner-magnetometer, similar to that described by Foster (1966), while most were measured on a 24 Hz fast spinner-magnetometer (Lumb & Roberts 1969).

The NRM directions for specimens from the same sample were averaged to give sample NRMs. The sample NRMs for each site were then averaged to give site NRMs which are listed in Table 2 (not corrected for tectonic tilt). The Koenigsberger ratio (Qn), in Table 2, gives some indication of the stability of the remanent magnetisation (Irving 1964; McElhinny 1973). Most of the specimens have high to very high Koenigsberger ratios which indicate good stability. Exceptions are the thin-bedded rhyolite tuffs from sites 29–36, the dacite from sites 38 and 40, and one sample of andesite from site 7.

Demagnetisation

Specimens from representative sites were tested by stepwise demagnetisation by tumbling in an alternating field using a two-axis tumbler described by McElhinny (1966). The demagnetisation steps began at 5 mT, and were increased by intervals of 5 mT to a maximum level of 50 mT, or until specimens began to display instability.

Interpretation of stepwise demagnetisation depends on both direction and intensity changes, and a method of plotting directions and intensities together (Zijderveld 1967) is here used to illustrate four samples representative of dacite, rhyolite, ignimbrite, and andesite (see Fig. 4). In this method, the magnetisation vector at each demagnetisation step is plotted as two components, one in the horizontal plane, and the other in the vertical plane, both vector components being plotted on the one figure. When each set of vector components lies on a straight line that passes through the origin of the graph, it can be inferred that only one component of magnetisation is being removed, and that it is likely to be a primary component. Deviation from the straight line relationship can be expected at early steps of the demagnetisation if secondary components with low coercivity are being removed (e.g., Fig. 4C).

The results illustrated by Figs 2, 3, and 4 all indicate that the samples tested have strong stable primary magnetisations.

A few samples showed irregular direction changes in fields greater than 25 mT (Fig. 2). In some cases, this may have been caused by the magnetisation becoming unstable in strong fields; in others, by the intensity of magnetisation decreasing to near the measuring limits of the magnetometers.

The pitchstone from site 56 gave an anomalous NRM with a low and reversed inclination, but after demagnetisation, gave a normal direction with a high inclination. It was stable in fields of 25 mT and higher. A vector plot similar to Fig. 4 revealed a secondary component present up to 25 mT, and a primary component on demagnetisation in higher fields. For all specimens except those from site 56, 15 mT was chosen as the standard demagnetisation field.

Tectonic Tilt Corrections

New Zealand Mesozoic rocks are usually tectonically tilted, and estimation of the correct tectonic tilt provides a major constraint on the accuracy of the paleomagnetic site directions and calculated pole positions. Tectonic tilt is commonly difficult to measure in volcanic rocks, especially in the more viscous rhyolites, dacites, and andesites. Moreover, variations in tilt due to local differential block tilting make it generally impossible to use a regional tilt correction.

Banded agate amygdules in the andesite and dacite flows were formed when the flows were cooling, or very soon after, and many of them contain parallel banded centres (Fig. 5). The bands vary in thickness from microscopic to several centimetres, and are commonly separated from one another by thin films of a white amorphous mineral. In thin section, the bands can be seen to consist of intergrown radiating clusters of cryptocrystalline quartz nucleated on the base of each band. The bands were initially horizontal (Oliver 1977) and, therefore, act as reliable internal tiltmeters. The amygdules range in size from less than 1 mm to greater than 1 m. Where only small amygdules were present, several apparent dips were measured to determine the true dip and strike. Where the amygdules were

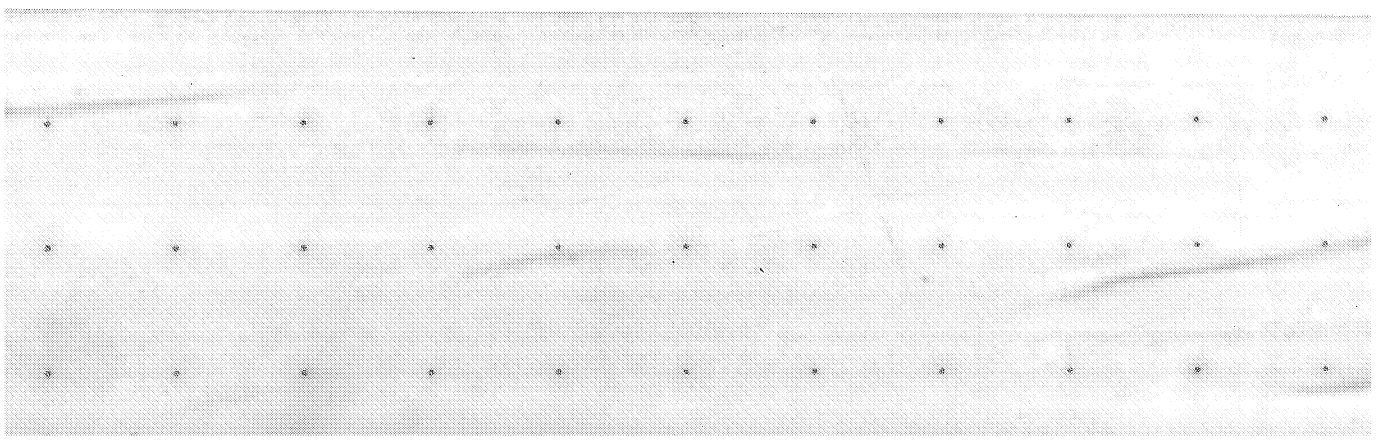


TABLE 1—Locations of paleomagnetic sample sites.

Site	Rock Type	Locality	Grid Ref *
1	Andesite	Ashburton Gorge	760424
2	Andesite	Ashburton Gorge	760425
3	Andesite	Ashburton Gorge	760423
4	Andesite	Ashburton Gorge	762425
5	Andesite	Ashburton Gorge	762427
6	Andesite	Ashburton Gorge	763429
7	Andesite	Ashburton Gorge	762432
8	Andesite	Ashburton Gorge	762435
9	Andesite	Ashburton Gorge	760437
10	Andesite	Ashburton Gorge	759429
11	Andesite	Ashburton Gorge	759427
12	Andesite	Blondin Stream	808398
13	Andesite	Blue Duck Stream	783450
14	Andesite	Blue Duck Stream	782450
15	Andesite	Blue Duck Stream	782450
16	Rhyolite	Blue Duck Stream	792448
17	Rhyolite	Burma Road	811437
18	Rhyolite	Morgan Stream	859483
19	Rhyolite	Woolshed Creek	848487
20	Rhyolite	Woolshed Creek	827460
21	Rhyolite	Mt Alford	984504
22	Pitchstone	Mt Alford	984504
23	Rhyolite	Blue Duck Stream	786448
24	Andesite	Morgan Stream	870483
25	Andesite	Morgan Stream	867485
26	Rhyolite Tuff	Morgan Stream	867486
27	Andesite	Morgan Stream	866486
28	Andesite	Woolshed Creek	848488
29	Rhyolite Tuff	Gawler Downs	884309
30	Rhyolite Tuff	Gawler Downs	884309
31	Rhyolite Tuff	Gawler Downs	884309
32	Rhyolite Tuff	Mt Alford	989501
33	Rhyolite Tuff	Mt Alford	984504
34	Rhyolite Tuff	Mt Alford	985504
35	Rhyolite Tuff	Waikare	811316
36	Rhyolite Tuff	Morgan Stream	867486
37	Dacite	Gawler Downs	883308
38	Dacite	Gawler Downs	882306
39	Dacite	Waikare	812314
40	Dacite	Gawler Downs	886298
41	Dacite	Hinds River	812374
61	Dolerite	Mt Alford	988501
42	Rhyolite Tuff	Mt Alford	989501
43	Dolerite	Mt Alford	985504
44	Dolerite	Mt Alford	985504
45	Rhyolite	Rangitata Gorge	602348
46	Rhyolite	Rangitata Gorge	571361
47	Rhyolite	Rangitata Gorge	587357
48	Andesite	Rangitata Gorge	647327
49	Andesite	Rangitata Gorge	647326
50	Andesite	Rangitata Gorge	645329
51	Andesite	Rangitata Gorge	577349
52	Andesite	Rangitata Gorge	635337
53	Ignimbrite	Rakaia Gorge	S82/125581
54	Ignimbrite	Round Top	S74/166605
55	Ignimbrite	Round Top	S74/169608
56	Pitchstone	Round Top	S74/169606
57	Ignimbrite	Round Top	S74/168605
58	Rhyolite	Mt Misery	S74/327645
59	Andesite	Round Top	S74/170609
60	Basalt	Pullwool Peak Road	S74/278636

*Grid Reference = NZMS 1 Sheet S81, unless otherwise indicated.

TABLE 2—Rock magnetic properties of Mt Somers Volcanics.

Site	N	J	D	I	χ	Qn
1	3	3.68	47	-85	0.65	120
2	1	0.249	171	-81	0.52	10
3	1	2.63	93	-80	1.3	43
4	1	0.328	176	-83	2.3	3.0
5	1	7.66	125	-71	2.3	70
6	1	1.64	122	-76	8.3	4.2
7	1	0.000659	212	-60	1.1	0.1
8	1	0.123	17	-81	1.9	1.3
9	1	0.123	64	-75	1.5	1.7
10	1	1.28	103	-78	0.77	3.5
11	1	9.20	156	-79	1.8	108
12	1	3.03	19	-71	16	4.0
13	1	2.13	50	-64	3.7	12
14	1	4.12	54	-67	8.9	9.8
15	1	2.98	68	-62	0.10	630
16	3	0.120	9	-79	3.1	0.8
17	2	0.138	22	-75	2.6	1.1
18	1	0.0565	198	-81	0.54	2.2
19	1	5.68	147	-71	3.5	34
20	1	0.135	39	-80	2.9	1.0
21	1	0.387	232	-66	3.9	2.1
22	1	2.66	219	-35	4.8	12
23	1	0.0534	91	-70	0.14	8.1
24	1	7.32	247	-71	1.3	119
25	1	0.701	205	-76	1.2	1.2
26	1	0.00390	205	-72	0.10	0.8
27	1	3.68	215	-50	1.9	41
28	1	4.56	104	-84	8.4	11
29	3	0.00256	322	-86	0.36	0.2
30	2	0.00290	1	-81	0.55	0.1
31	2	0.00325	241	-79	0.60	0.1
32	1	0.142	185	-76	0.77	3.9
33	1	0.00184	246	-71	0.15	0.3
34	1	0.00156	257	-69	0.10	0.3
35	1	0.00104	36	-82	0.23	0.1
36*	1	-	-	-	-	-
37	1	1.71	306	-76	2.6	14
38	2	0.0172	342	-66	0.98	0.4
39	1	8.87	78	-62	8.3	23
40	1	0.00121	121	-56	0.64	0.1
41	1	0.213	21	-79	1.6	2.8
61	1	3.45	152	-72	4.8	15
42	1	0.0931	248	-74	0.19	10
43	1	3.11	190	-73	13	5.1
44	1	3.09	271	-74	19	3.4
45	2	0.433	0	-73	11	0.8
46	1	0.879	18	-63	22	0.8
47	2	0.420	19	-66	15	0.6
48	2	10.3	355	-76	19	11
49	1	6.27	6	-64	9.5	14
50	1	5.18	354	-77	28	3.9
51	3	0.264	56	-75	5.6	1.0
52	1	6.52	326	-28	5.3	26
53	2	1.61	323	-44	3.3	10
54	2	0.533	351	-82	1.0	11
55*	1	-	-	-	-	-
56	1	3.30	110	+22	1.6	44
57	1	0.703	163	-83	2.0	7.5
58	1	0.00440	18	-67	0.1	0.9
59	1	1.46	32	-76	4.0	7.7
60	1	0.0292	335	-66	1.9	0.3

N = number of samples from each site
J = intensity of natural remanent magnetisation (A/m)
D = declination of natural remanent magnetisation (Degrees)
I = inclination of natural remanent magnetisation (Degrees)
χ = volume magnetic susceptibility (S.I. units $\times 10^3$)
Qn = Koenigsberger Ratio ($J/\chi H$); where H = 47·34 A/m (59·5 μT) the intensity of the geomagnetic field at Mt Somers
* = samples too weak for measurement

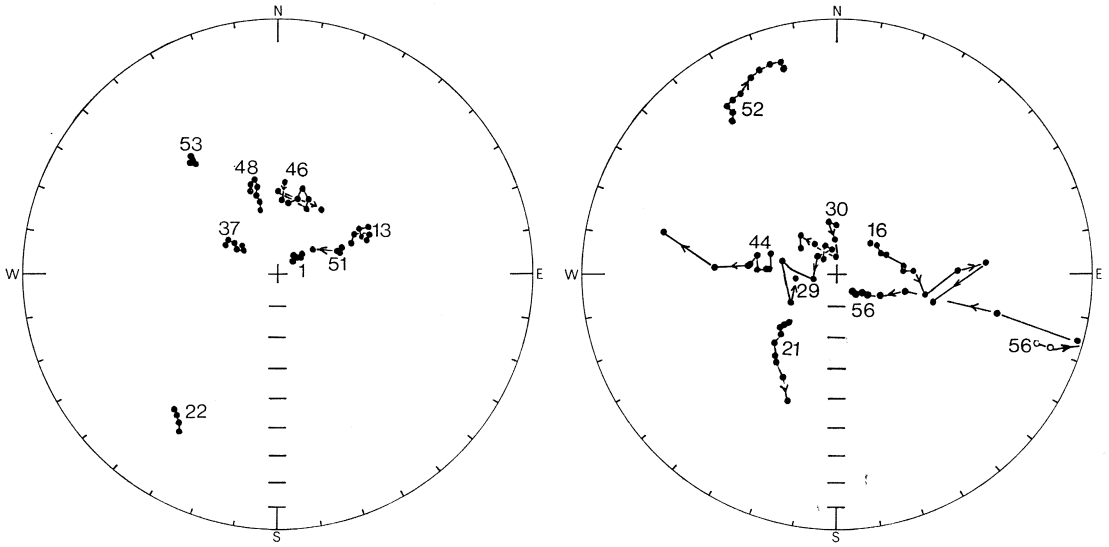


FIG. 2—Changes in direction of magnetisation on progressive AF demagnetisation. Numbers refer to samples from sites shown on Fig. 1.

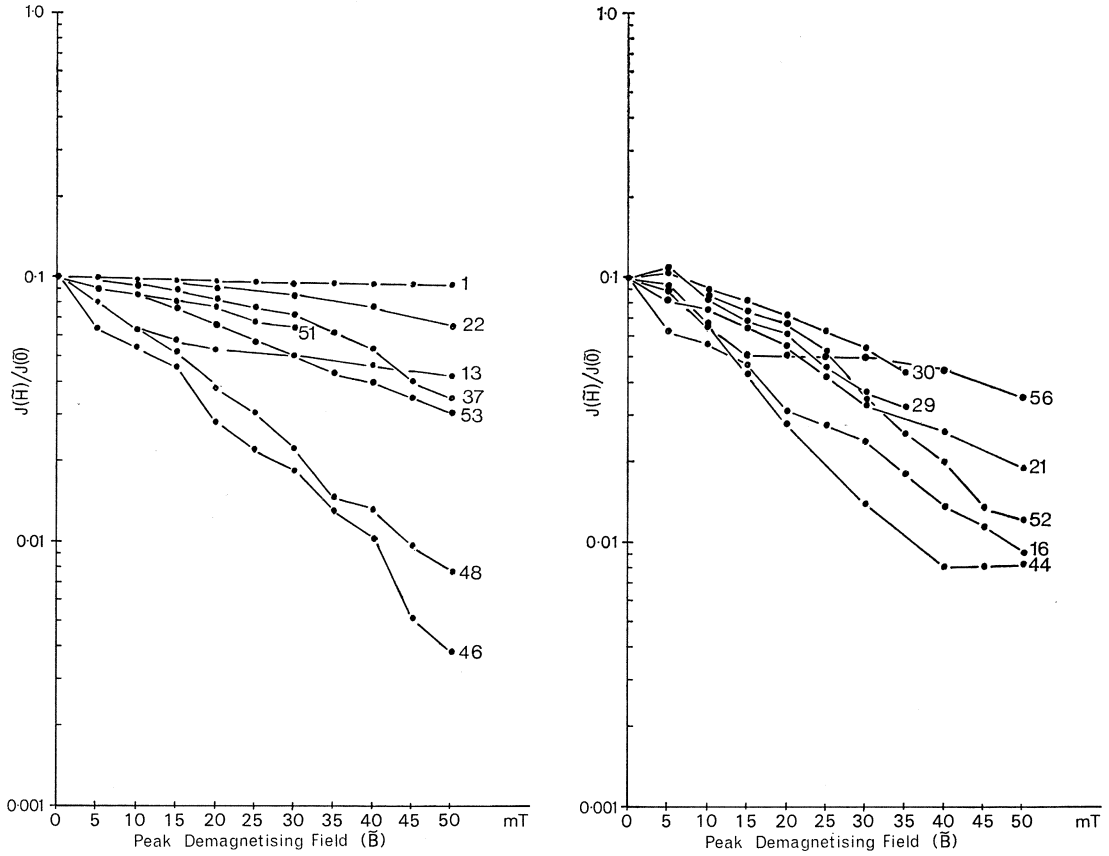


FIG. 3—Normalised intensities of magnetisation on progressive AF demagnetisation. Numbers refer to samples from sites sh own on Fig. 1.

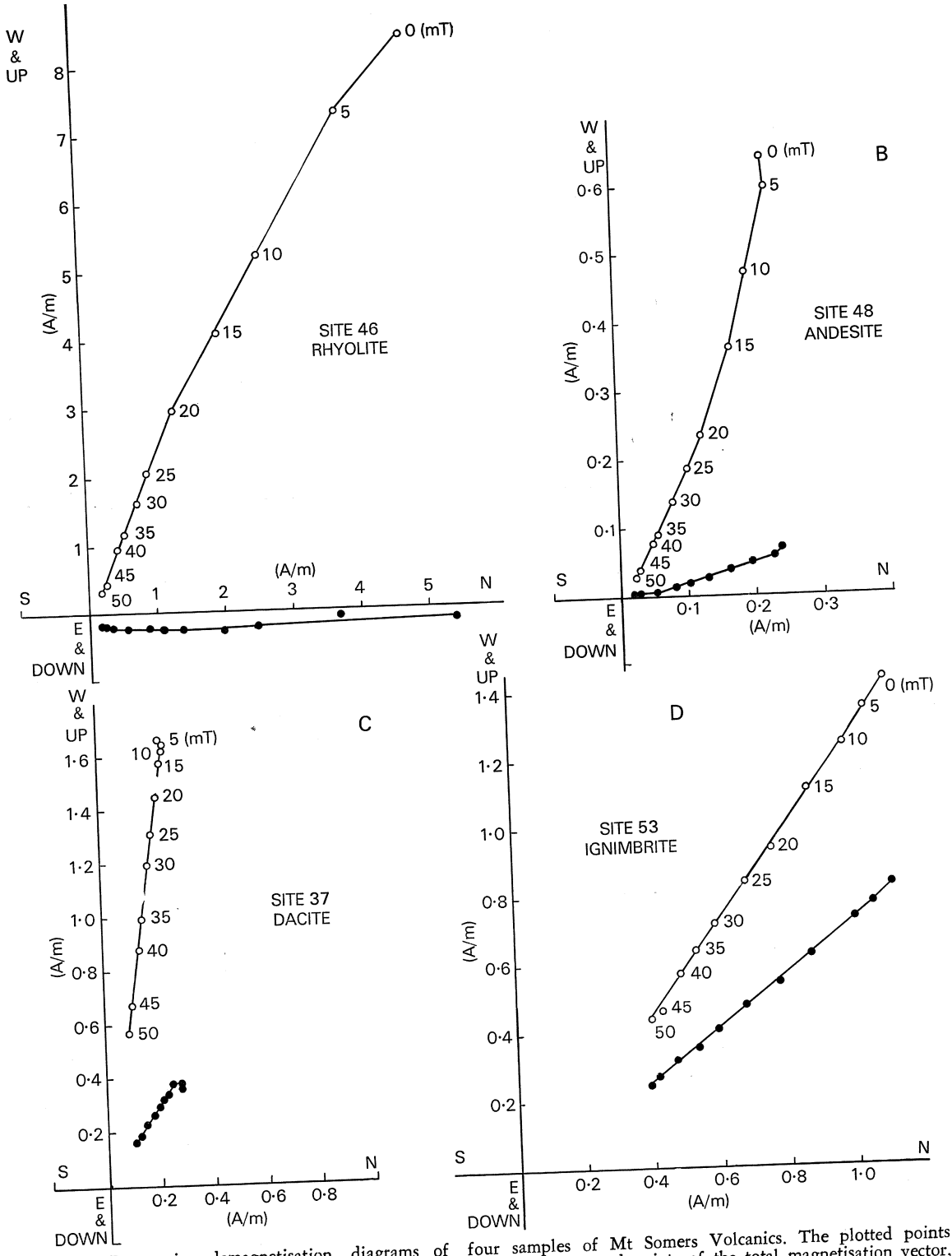


FIG. 4—Progressive demagnetisation diagrams of four samples of Mt Somers Volcanics. The plotted points represent successive positions of the orthogonal projections of the end points of the total magnetisation vector, on progressive demagnetisation. Solid circles denote the projections on the horizontal N, S, E, W plane; open circles, the projections on the vertical N, S, Up, Down plane. Peak AC demagnetising fields are shown in mT.



FIG. 5—Andesite with banded agate amygdules, Morgan Stream.

larger, the agates could usually be parted along the planes between layers, and the dip measured directly by placing a compass on the plane surface.

The rhyolite tuffs have sub-aqueous sedimentary structures, including good graded bedding, and were laid down in shallow lakes indicated by interbedded carbonaceous silts. The attitude of the beds, especially those of coarser grade that have undergone little compaction, provides a reliable measure of tectonic tilt.

Large-scale regular parallel banding in ignimbrite is assumed to have been horizontal except in flow-folded rheo-ignimbrite (Rittmann 1962) such as that sampled from Round Top in the Malvern Hills area, which has locally variable banding. With the flow-folded ignimbrite, a reversed "fold-test" showed that the cleaned magnetic directions have good grouping, but not when tilt corrections based on the flow-banding are applied, indicating the folding of the flow-banding occurred prior to cooling of the ignimbrite.

The rhyolite domes are the most difficult volcanics on which to determine a tectonic tilt as there are no internal features to indicate the original attitude. The tilt corrections were, therefore, extrapolated from the tilt of nearby overlying Lower Paleocene coal measures, or from that of the underlying andesite.

The NRM and demagnetised directions corrected for tectonic tilt are given in Table 3. Samples were eliminated that were unstable on demagnetisation, or where the tectonic correction data were ambiguous.

Site 26 was a baked porcellaneous rhyolite tuff overlain by andesite. This tuff bed has a dip of 75° S and a strike of 265° which, when applied as a tectonic correction, gave a magnetic declination of 343° and an inclination of −30°. When the tilt correction for the overlying andesite was applied to the tuff, the direction was found to be similar to that from the andesite (site 25, Table 3). This baked contact test confirms that the magnetism of the tuff bed was acquired after the tilting of the bed and at the time of extrusion of the andesite. The similarity of the magnetic directions, after cleaning, of the baked tuff and the andesite, also shows that both measurements are of stable thermoremanent magnetisation (TRM) and, as the andesite direction did not change appreciably on cleaning, that the NRM of the andesite is also the primary TRM. This result confirms the interpretation, based on the demagnetisation tests, that the magnetic remanence of the andesite is a single, stable, primary component magnetisation.

Plots of NRM directions and directions after cleaning, corrected for tectonic tilt (Fig. 6), group well. The directions after cleaning show more scatter than the uncleaned NRM directions, but the mean direction is similar for both. The statistics (Table 4) imply the grouping is better in the cleaned samples, but this is due to the presence of the reversed NRM of site 56 in the uncleaned group. If site 56 is excluded, the mean uncleaned NRM is D = 348°, I = −83·6 (N = 45, K = 41·4, α_{95} = 3·3, R = 43·9) which is in agreement with the visual grouping in Fig. 6.

Paleomagnetic Pole Position

When the magnetic inclination is very steep, the intensity of the declination component is weak and difficult to measure. Small errors in sample orientation while coring and while measuring in the magnetometer, combined with errors in the estimation of tectonic tilt, have an exaggerated effect on the declination direction determined. The observed spread of magnetic declinations (Fig. 6), and the calculated virtual geomagnetic pole positions may, therefore, be partly due to the high inclinations. In spite of these factors, the directions obtained from the Mount Somers Volcanics are remarkably consistent, giving a statistically significant mean direction at the 95% level of confidence (Table 4). Much of the spread in the results from the Mount Somers Volcanics can doubtless be attributed to

TABLE 3—Directions of magnetisation corrected for tectonic tilt

Site No.	Tectonic Tilt Correction		Corrected Directions			
			NRM		Cleaned*	
	Strike	Dip	D	I	D	I
1	24	6 E	344	-83	345	-85
2	24	6 E	215	-82	195	-86
3	24	6 E	70	-86	107	-85
4	360	10 E	233	-78	231	-77
5	360	10 E	150	-78	150	-74
6	360	10 E	177	-82	130	-82
8	360	10 E	320	-79	331	-78
9	340	15 E	340	-89	338	-89
10	340	10 E	155	-83	149	-83
11	24	6 E	195	-83	188	-82
12	80	15 N	71	-81	51	-80
13	330	35 E	264	-80	261	-84
14	340	30 E	288	-80	342	-83
15	340	30 E	265	-88	263	-88
16		0	9	-79	32	-79
17	221	13 SE	346	-68	352	-66
18	105	35 S	14	-64	0	-66
19	80	25 S	37	-80	45	-81
20	36	16 W	358	-75	85	-65
21	310	25 SW	341	-85	270	-84
24	105	20 S	315	-73	309	-73
25	105	35 S	2	-67	41	-75
26	105	35 S	6	-73	9	-76
27	105	35 S	254	-76	264	-77
28	100	35 S	21	-65	22	-55
29	182	16 E	280	-71	282	-67
30	138	9 NE	298	-83	245	-82
31	269	11 S	297	-79	289	-77
32	90	10 S	203	-86	194	-80
33		0	246	-71	254	-75
34	310	24 SW	340	-76	347	-72
35	295	19 S	28	-63	35	-74
41		0	21	-79	18	-79
61	90	10 S	122	-80	162	-70
42	110	15 S	302	-78	273	-78
43	320	15 SW	130	-79	148	-79
44	320	15 SW	336	-79	322	-74
48		0	355	-76	352	-71
49		0	6	-64	11	-63
50			354	-77	344	-71
51	209	21 E	349	-70	349	-70
53	40	45 W	46	-80	29	-80
54		0	351	-83	329	-82
56		0	110	+22	124	-81
57		0	163	-83	176	-71
59		0	32	-76	34	-77

*All samples cleaned in AC fields of 15 mT, except sample from site 56 which was cleaned in AC field of 30 mT.

secular variation, which is normally responsible for directional variations up to 20° from the axial dipole (Tarling 1971).

TECTONIC INTERPRETATION

Oroclinal Bending

The Mount Somers Volcanics unconformably overlie the sedimentary rocks of the Torlesse Supergroup which, in this area, from fossil evidence, are divided into an older Middle to Upper Triassic group, and a younger Upper Jurassic (Puaroan) group (Oliver 1977). The belt of Upper Jurassic rocks forms a synclinal fault-bounded strip within the older Triassic rocks that follows an arcuate trend around the inner margin of the Canterbury Plains. The same arcuate trend is shown by the plotted strike directions of the bedding in both the Jurassic and Triassic rocks (Fig. 7). The megascopic folding of the Torlesse rocks appears to predate the block-faulted Mount Somers Volcanics and to postdate the youngest folded Jurassic beds. The arcuate trend of the Torlesse rocks appears to follow the Early Cretaceous (Rangitata) fold belt (being part of the regional swing in strike of the Haast Schist and Torlesse metagreywackes, from North Otago–South Canterbury (NW–SE) to mid–North Canterbury (NE–SW)) that is generally attributed to oroclinal bending of the New Zealand Geosyncline in the Rangitata Orogeny (Fleming 1970; Grindley 1974). The paleomagnetic data support the field evidence of pre-Upper Cretaceous folding (and bending), since no systematic change in the mean magnetic directions has been observed across the area as would be expected if bending had taken place later than the Early Cretaceous.

The paleomagnetic data do not resolve the problem of when the dextral displacement on the Alpine Fault took place, as the Mount Somers area is 75 km east of the Alpine Fault and outside the area directly affected by dextral shear at the present day. However, if any dextral shear accompanied oroclinal bending, then this also would be of Early Cretaceous age. It is possible that dextral displacement along the Alpine Fault has been episodic, with movement in both the Rangitata and Kaikoura Orogenies, as suggested by Grindley (1963; 1974) and Suggate (1963), although the relative magnitudes of these movements cannot yet be determined.

Polar-Wander Paths

The Cretaceous–Cenozoic polar-wander path for Australia has been established (McElhinny & Wellman 1969; Wellman *et al.* 1969) and further refined (McElhinny *et al.* 1974) from studies of eastern Australian Cenozoic volcanic rocks, the ages of which have been precisely determined by K-Ar dating (Wellman & McDougall 1974a, b). The paleomagnetic results have been divided into six Cenozoic age groups ranging from 60 Ma B.P. (Paleocene) to the present, and a seventh early Late Cretaceous age group. This latter result combines the results from the Cygnet alkaline complex of Tasmania (Robertson & Hastie 1961), the Mt Dromedary alkaline complex of New South Wales (Robertson 1963), and the Bunbury basalts of southwest Australia (Wellman & McDougall 1974a; Schmidt & Embleton 1976). The apparent polar-wander path (APWP) for Australia, as determined by McElhinny *et al.* (1974) is reproduced here (Fig. 8). Generally, this Australian APWP mirrors the northward drift of Australia from Antarctica during the Cenozoic, but shows pronounced westward excursions from a direct path between 15 and 25 Ma B.P. and between 25 and 30 Ma B.P.

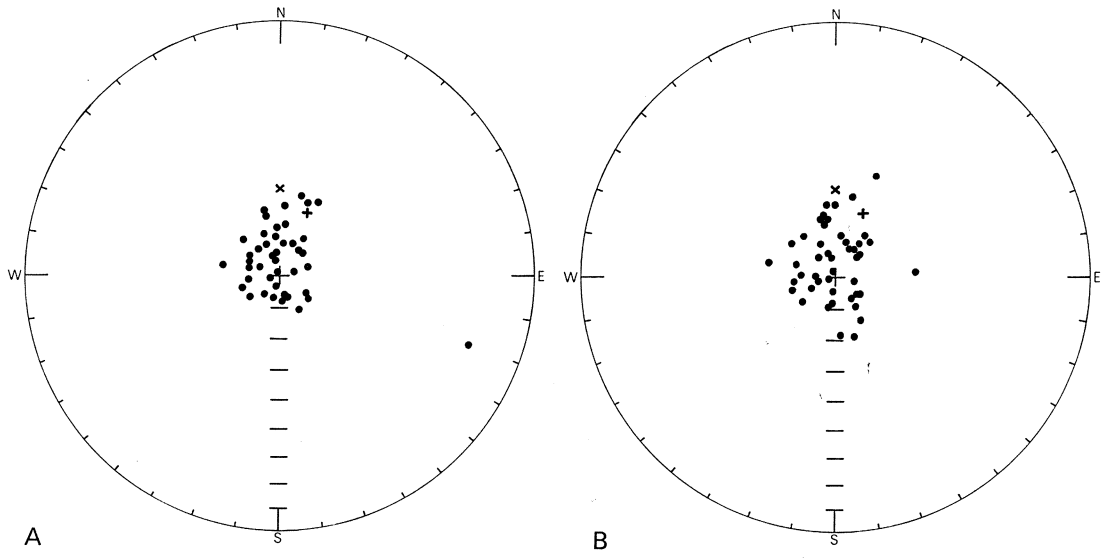


FIG. 6—Directions of magnetisation of cleaned samples corrected for tectonic tilt. Axial dipole field direction (×), and present day field (+) at Mount Somers. A—NR M directions; B—directions after AF demagnetisation.

TABLE 4—Mean magnetic directions and paleomagnetic pole position for the Mt Somers Volcanics.

Samples	Mean magnetic directions Declination	Inclination	N	K	α_{95}	R
Uncleaned	358	−84	46	18·0	5·1	43·5
Cleaned	354	−85	46	31·7	3·8	44·6

Samples	Paleomagnetic pole position Latitude	Longitude	N	K	α_{95}	R
Uncleaned	54° S	174° E	46	9·4	7·2	41·2
Cleaned	52° S	174° E	46	10·0	7·0	41·5

N = Number of sites
K = Fisher's precision parameter
α_{95} = Radius of 95% cone of confidence of mean value
R = Resultant magnitude of sum of unit vectors

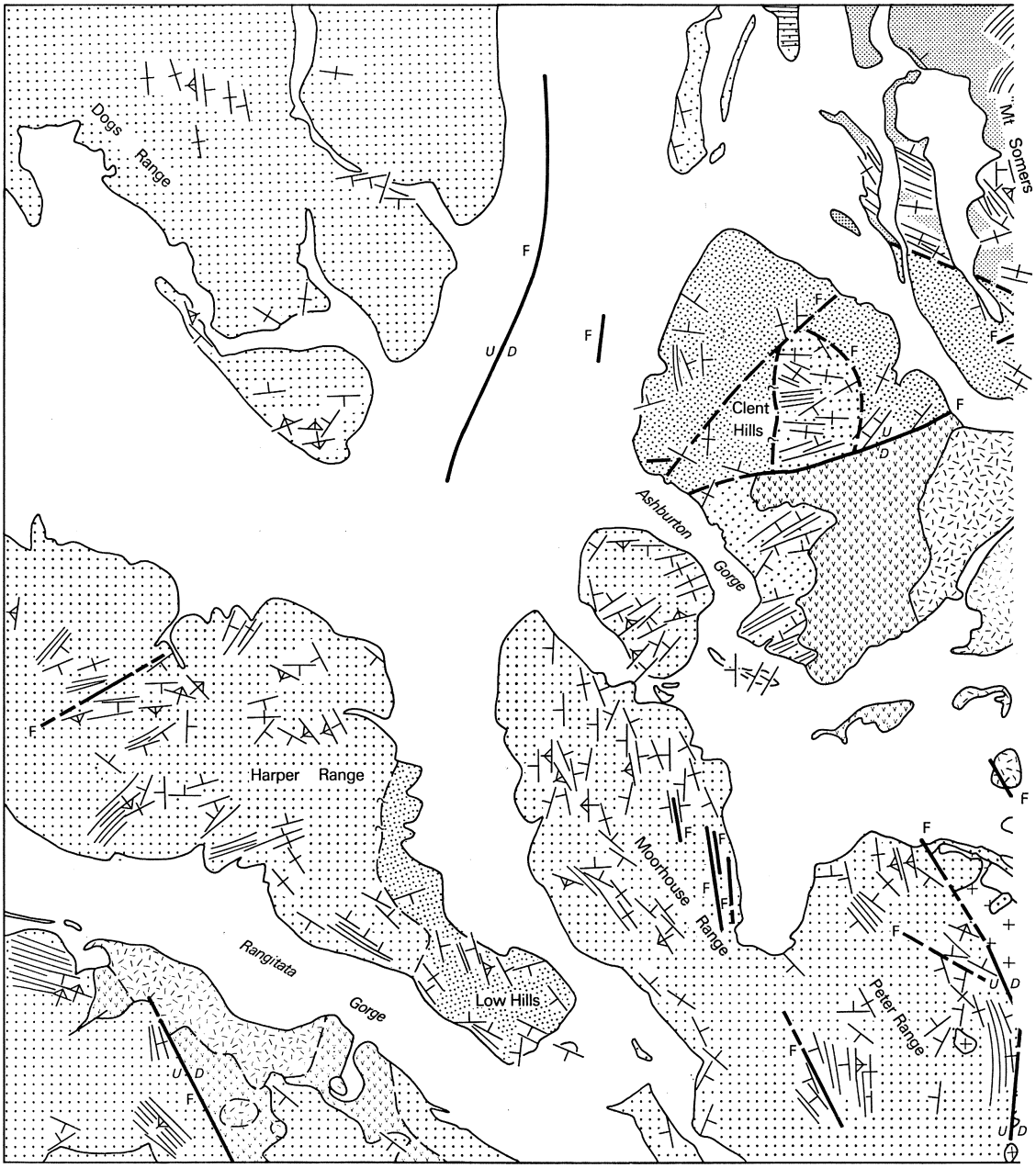


FIG. 7—Diagram showing structural trends of folds and bedding in the Mesozoic basement rocks of the Mt Somers area (after Oliver 1977).

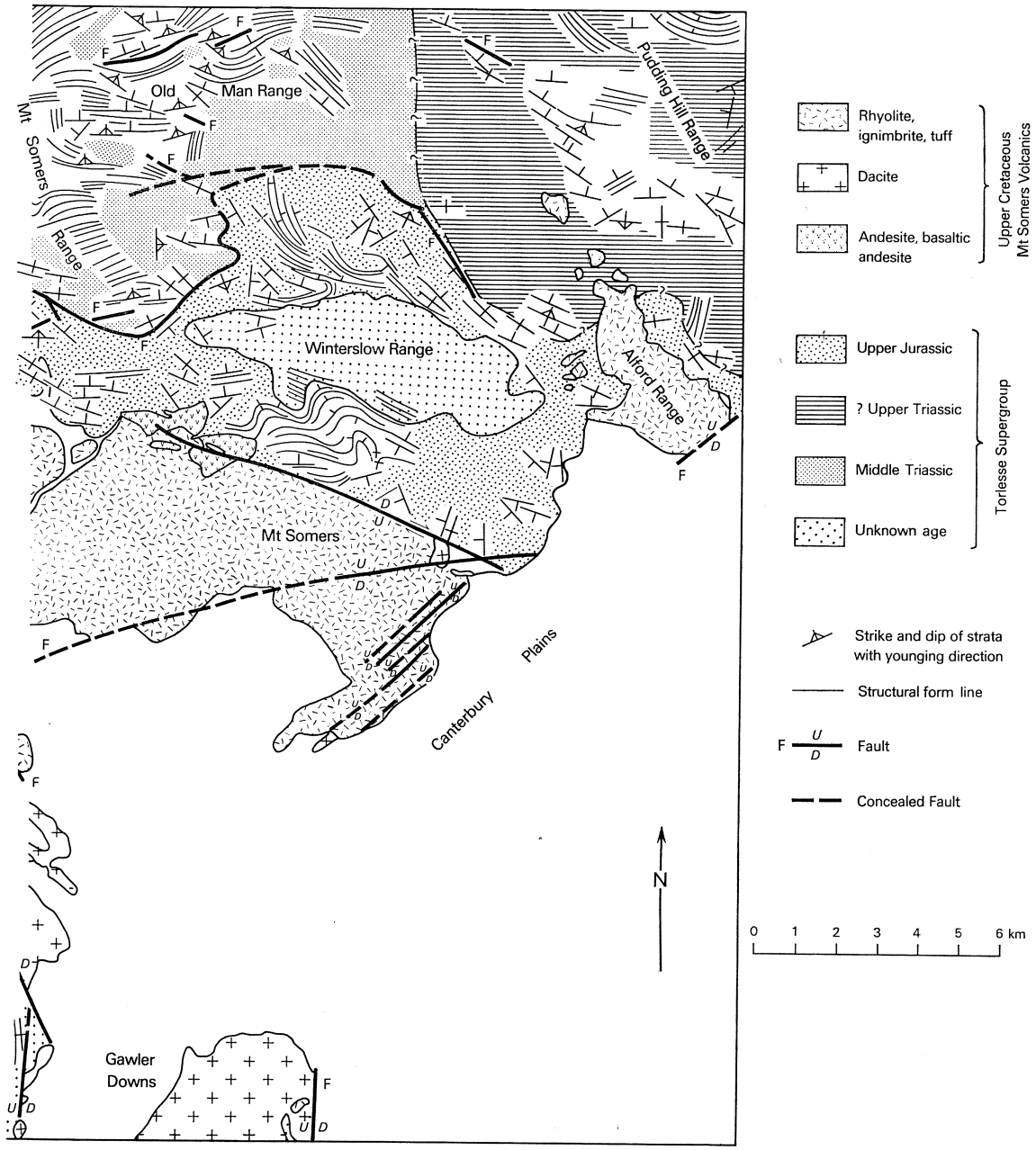


FIG. 7—(*Continued*)

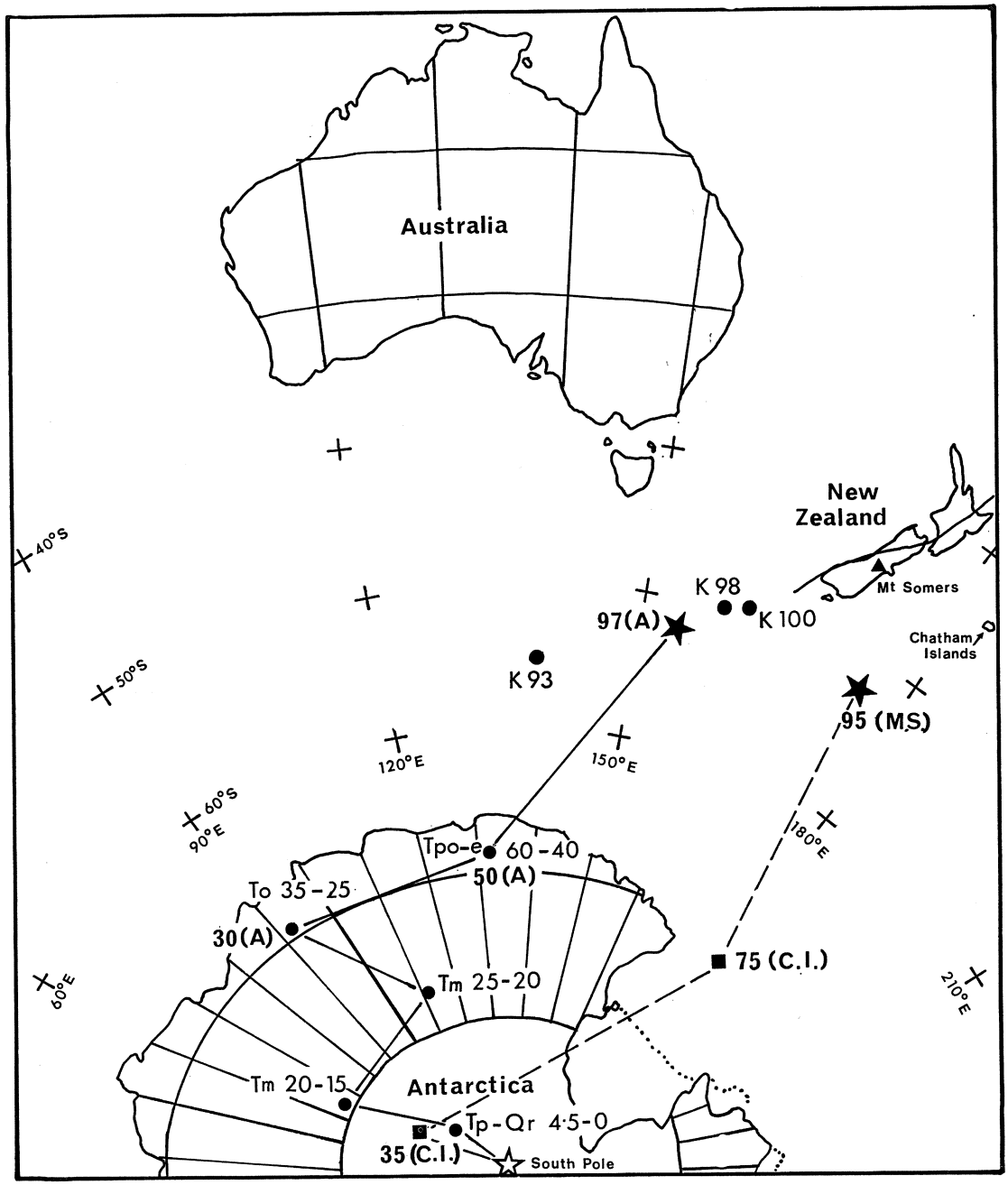


FIG. 8—Australian apparent Polar-Wander Path (McElhinny *et al.* 1974), compared with New Zealand apparent Polar-Wander path inferred from Chatham Islands (Grindley *et al.* 1977) and Mt Somers (this paper).

An apparent polar-wander path for New Zealand can be constructed from the Chatham Islands data (Grindley *et al.* 1977) and results from Mount Somers, and lies some 1000–5000 km east of the Australian APWP. The differences correspond to the finite rotational displacements across the Indian-Pacific and other plate boundaries (such as the Tasman spreading axis) over the past 90 Ma. Finite poles for these rotations lie on the great circles bisecting the lines joining age-equivalent positions on the separate paths (Fig. 8).

When the continents are plotted on a pre-drift 95 Ma reconstruction, such as the computer plot of Crook & Belbin (1975; 1978), the Mount Somers pole position falls close to the 95 Ma mean Australian pole (Grindley & Davey in press). Provided that post-95 Ma oroclinal bending is less than about 45°, the circles of confidence of the Australian (7°) and New Zealand (7°) paleopoles will overlap. Plate tectonic reconstructions of Molnar *et al.* (1975) and Wellman (1975; 1976) incorporate Cenozoic oroclinal bending of the New Zealand Geosyncline of between 80° and 100°, an amount of bending incompatible with the paleomagnetic data, despite the relatively high inclinations. Further paleomagnetic work, especially west of the Alpine Fault and closer to the Alpine Fault on the east, is required to resolve the problem of oroclinal bending in relation to plate movements.

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